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AUTOMATIC EXAM CORRECTION FRAMEWORK FOR SHORT WRITTEN ANSWERS USING NLP

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ABSTRACT

Automatic grading requires the adaption of the latest technologies. It has become essential especially when most of the courses became online courses. The traditional evaluation system involves a particular evaluator of a particular subject to manually evaluate the answers written by students. It is a very time consuming process for an evaluator to evaluate answer scripts of each and every student manually. Hence an automatic exam paper evaluation framework has been proposed to enable automatic correction of answers and grading system. This system can be used in all educational institutions for reducing the work of manual answer scripts evaluation. The proposed automatic exam correction framework for short written answers [AECF] has been developed using tools like OCR, Python programming languages and packages in python like NLTK. Python flask server has been used for backend and UI has been enhanced using HTML and CSS.

Keywords: AECF, OCR, NLTK.

I. INTRODUCTION

Nowadays many people use images to represent and symbolize information by convertingan image into digital form where different operations are performed on it for the extraction of information though the techniques of image processing. The important information that is extracted from images is more popular. Image recognition can be a vital analysis half for wide applications. Within the comparative pattern recognition field of pc, the one difficulttask is actual automatic recognition of individual human handwriting. Even though, the variances do not origin any struggle to human. Yet, but it is further complicated to instruct computers towards recognize wide-ranging handwriting. Intended for image recognition difficulty such since handwritten categorization, it is extremely significant to create away how information or data are represent in images.

II. LITERATURE SURVEY

A literature survey or a literature review in a project report shows the various analyses and research made in the field of interest and the results already published, taking into account thevarious parameters of the project and the extent of the project. Literature survey is mainly carried out in order to analyze the background of the current project which helps to find out flaws in the existing system & guides on which unsolved problems we can work out. So, the following topics not only illustrate the background of the project but also uncover the problems and flaws which motivated to propose solutions and work on this project.

Here are several proposed approaches and systems in the field of automatic scoring and grading. Most of them worked on the grading of essays and short answers.

Hossam Magdy Balaha and Mahmoud M. Saafan. [1] suggested Text semantic similarity, Equations Matching, MCQs, the advantage is Evaluation of MCQs and they didn't evaluate writtenanswer scripts.

Hassan et al. [2] suggested a supervised learning algorithm for the short answer automatic scoring which was based on paragraph embeddings. They discussed significant deep learning-based models for generating paragraph embeddings. They presented a detailed choice criterion of paragraph embedding.

Pribadi et al. [3] developed a system for automatic short answer scoring. They compared multiple methods that applied the overlapping methods to determine the similarity degree between the references and students' answers. They showed that the cosine coefficient method reported the best results than the Dice and Jaccard coefficient methods.

Mohler et al. [4] investigated unsupervised techniques for the automatic short answer grading problem. They

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compared a set of text similarity knowledge-based and corpus-based measures. They reported the effect of domain and size on the corpus-based measures. They introduced a technique to improve system performance by integrating automatic feedback from the student answers.

Table No: 1 Brief Summary of Literature Survey

WORK	YEAR	DETAILS	ADVANTAGES	DISADVANTAGES
HossamMagdy & MD. Saafan .[1]	2021	Text semantic similarity, Equations Matching, MCQs	Evaluation of MCQs	Didn't evaluate written answer scripts
Hassan etal. [2]	2018	Generating Paragraph Embeddings	Supervised learning algorithm & deep learning based models	Didn't propose a general framework tofollow
Pribadi etal. [3]	2017	Evaluation of Essay and Similarity scores	They worked on essay & reported a set of similarity scores	They didn't work on Evaluating Written answers
Mohler etal. [4]	2011	Datasets were made accessible by Public	They compiled their dataset and made it for public usage	They didn't work on Evaluating Written answers
Mohler etal. [5]	2009	Implementation of unsupervised learning algorithms for text similarity	They approached unsupervised learning algorithms & compared a set of text similarity knowledge – based & corpus-based measures	Didn't work on equations nor MCQs.They didn't comparetheir work with the pre-built embedding models

III. SYSTEM ARCHITECTURE

The proposed architecture shown figure 1 consists of four layers. The first layer is responsible for accepting the student's answers to different questions. The second layer is responsible for combining each student's answer with the corresponding reference answer. The reference answer for a short answer can be keywords or the actual answer. The Third layer is the Auto Correction layer it checks the question type and determines if the question is a single step or multiple steps. The instructor should pass on each student's answer (with its correction) anonymously. The fourth layer is responsible for storing the results in the storage medium (i.e. database or cloud storage) and displaying the final report to the user. It does not generate new headers.

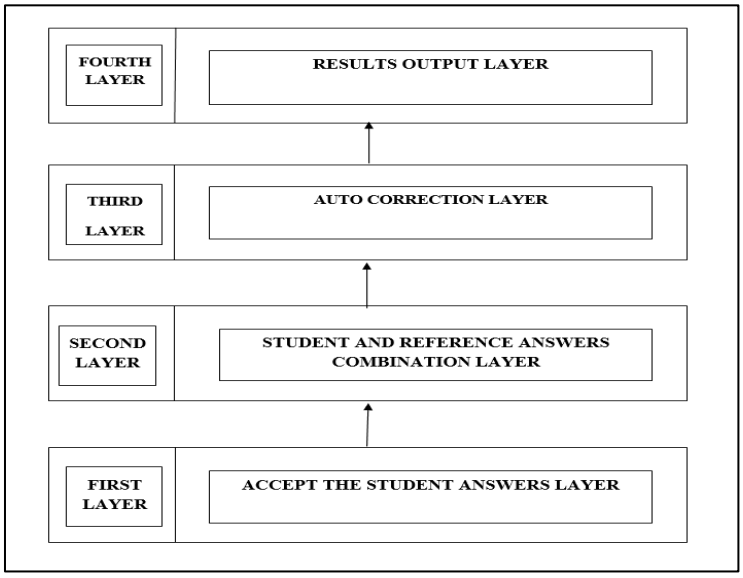

Figure 1: System Architecture of AECF

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Layer 1

The first layer is responsible for accepting the student's answers. This layer mainly consists offour components Student id, Question id, Question and student answer image as shown in figure 2.

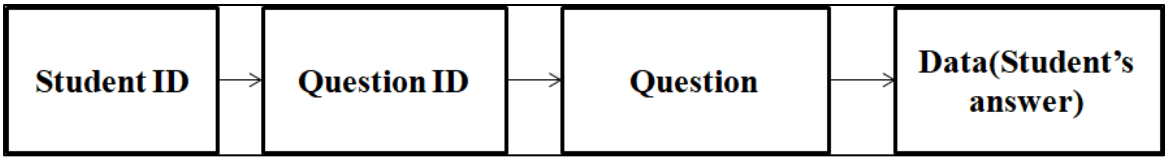

Figure 2: Layer 1 of AECF

The first layer accepts the student answer, As the student select the question ID a question is displayed student can upload the written answer image for that question. Once a student submits the answer, all the data like question id, student answer image is stored in MySQL database. For each and every question the student need to do same procedure to upload the answer.

Layer 2

The second layer is responsible for combining each student's answer with the corresponding reference answer. The reference answers are the answer typed by evaluator while assigning question as shown figure 3.

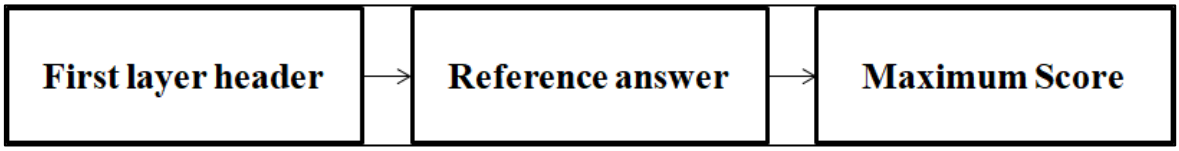

Figure 3: Layer 2 of AECF

The Second layer takes the data of first layer and reference answers are combined with the first layer header then the question maximum marks attached.

Layer 3

The Third layer is the Auto Correction layer it checks the extracts text from the image using OCR model and evaluates the student answer using NLP as shown in figure 4.

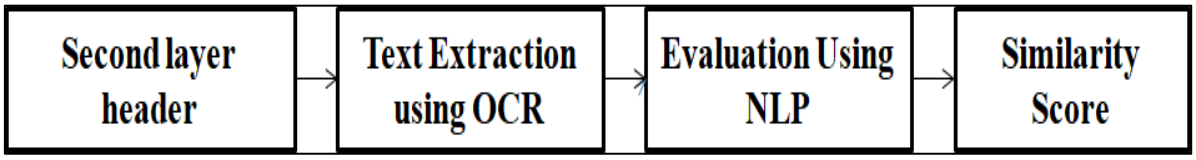

Figure 4: Layer 3 of AECF

In layer 3 the text is extracted from the image which is uploaded by the student. The extracted text and reference answers are compared using NLP for Evaluation. After Comparison the similarity score is generated.

Layer 4

The fourth layer is responsible for storing the results in the storage medium (i.e. database) and displaying the final results to the user. It does not generate new headers as shown in figure 5 below.

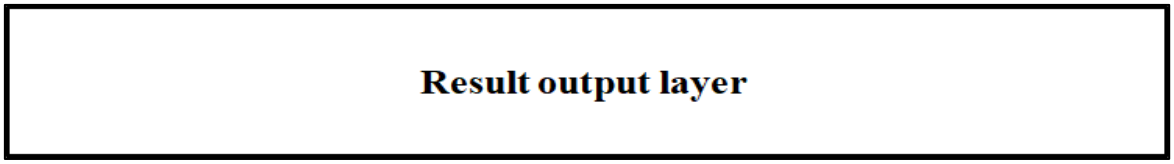

Figure 5: Layer 4 of AECF

The Automatic Exam Correction Framework [AECF] for Short Written Answers using NLP performs automatic evaluation of answer scripts that are written, scanned and uploaded in the form of images. The following are the methodologies used in this proposed system.

IV. SYSTEM MODULES

A. STUDENT MODULE

The Student First Need to register With the Required Credentials like USN, name, email etc. Then student need

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to login with username and password. The Username and password is verified with database if Credentials are correct it directs to answer uploading page. In Answer uploading page student must select the question number and student may either type or upload answer images. After Uploading the student can see the status uploaded as shown in figure 6.

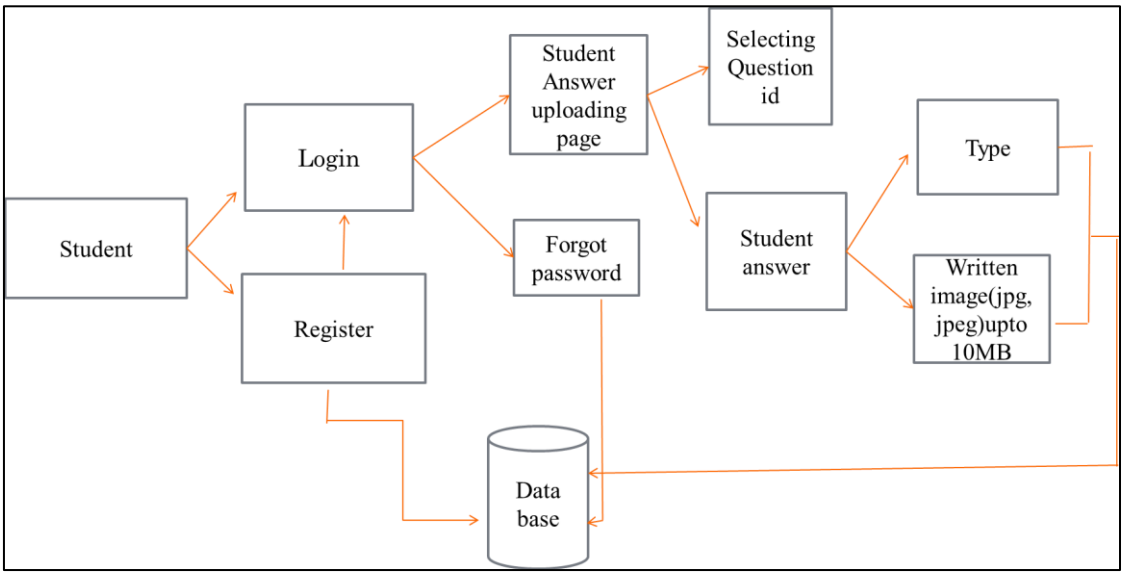

Figure 6: Student Module of AECF

B. STUDENT REGISTRATION MODULE

The student need to fill the details which are mandatory to register. The details are Name, USN, email, phone number, password. The Name and USN are primary keys which are need to be unique. If anyone field is not filled the system pops a instruction to fill the details, In order to register the student has to fill all the details compulsory. While registering the proper mail id must be given so that if user forgot password the password can be reset using the OTP sent to registered mail id. The password need to be strong with minimum one uppercase, one special character. After giving all the credentials the user can register as shown in figure 7 below.

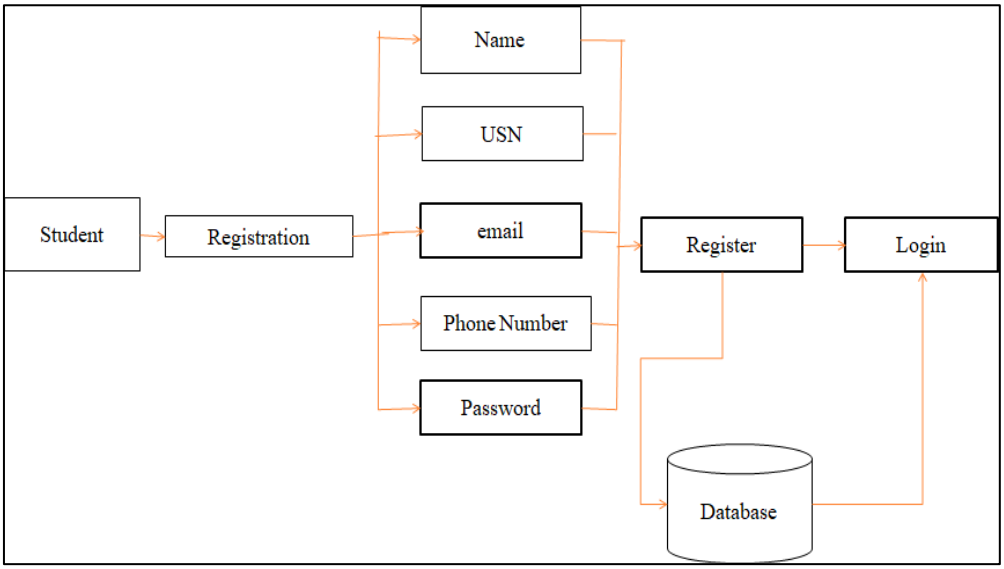

Figure 7: Student Registration of AECF

C. EVALUATOR MODULE

In this module Evaluator can login to our application using their credentials and they can addthe questions and answer keys. Evaluator can publish the results to the students as shown in figure 8 below.

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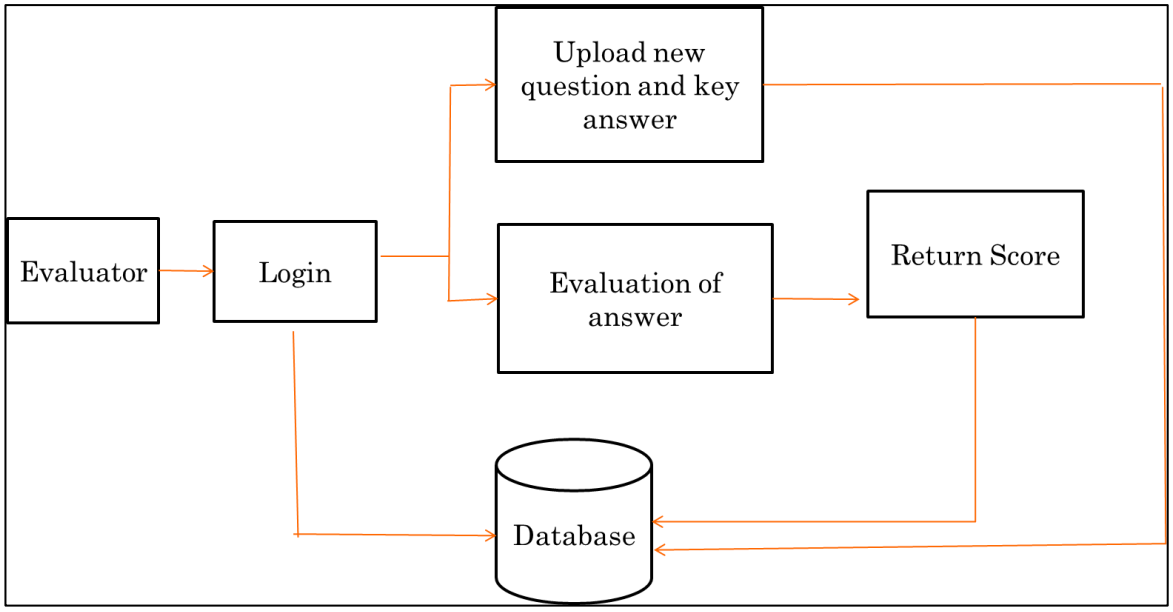

Figure 8: Evaluator Module of AECF

D. EVALUATION OF ANSWERS

This model will work when teachers click evaluate. All the answers are stored in our server in image format. In this module we are applying the OCR model to extract the content from the image. Once the text content is extracted we are applying the NLP tokenizing and vectorizing techniques to convert the text to vectors. Then we are comparing the cosine similarity between the actual answer with student answer as shown in figure 9 below. .

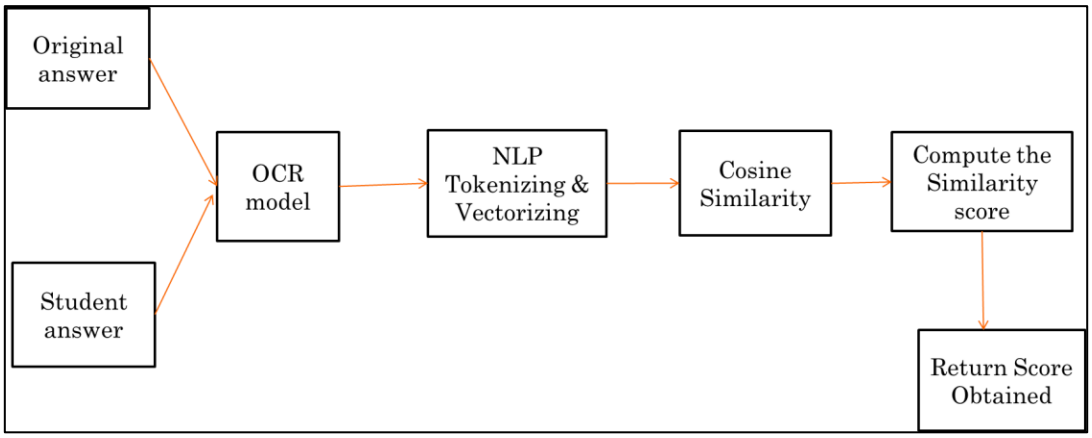

Figure 9: Evaluation of Answer of AECF

V. IMPLEMENTATION

OCR

In OCR processing, the scanned-in image or bitmap is analyzed for light and dark areas in order to identify each alphabetic letter or numeric digit. When a character is recognized, it is converted into an ASCII code. Special circuit boards and computer chips designed expressly for OCR are used to speed up the recognition process.

// Input : Query Image File

// Output: Text

STEP 1 : START

STEP 2 : Fetch the Uploaded files from database of particular question.

STEP 3 : CALL OCR module to extract text

LOOP process // to extract text character by character

STEP 4 : SAVE extracted text to temporary location

STEP 5 : CALL matching module

STEP 6 : if (Match found) then

DISPLAY results of matches

STEP 7 : end if

STEP 8 : return extracted text

NLP

Natural Language Processing is used for evaluating answers. The extracted text which is obtained from the OCR model is evaluated using NLP through multiple steps. First, the input sentence or the extracted text is broken down into smaller chunks such as words or sentence known as tokenization. Syntactic analysis or parsing is the next step in NLP which is used to draw exact meaning from the text. Semantic analysis helps in understanding the context of the text. Cosine similarity is a technique that is used to calculate similarity score of the text. The output is then obtained in the form of scores as shown below.

Algorithm Evaluation _of_answer(Original_ans, Student ans)

//Input : Original answer and Student answer images.

//Output : Actual score obtained.

Step 1 : Extract the text from the images.

//Here we are applying the Optical Character Recognization (OCR) model to extract the content from the images.

Step 2 : Call NLP functiona.

//Here after extracting the text we are applying the NLP tokenizing and vectorising techniques to convert text to vectors.

Step 3 : Calculate the Cosine Similarity between two vectors.

Vector 1← Actual answer vector

Vector 2 ←Student answer vector

Step 4 : Compute the similarity score between the original answer and student answer vector.

Score_obt ← Cosine_similarity_score (vector 1,vector 2)

Step 5 : Return the score obtained by the student.

VI. RESULTS AND DISCUSSION

A. PERFORMANCE EVALUATION

In the below figure 10 we have compared the performances of existing and proposed system the existing system was having some 35% accuracy Z. Chen, H. Zhang, X. Zhang, and L. Zhao [18] and in proposed system we have tried to increase the accuracy of the system up to 85%. And time constrained is also a very important factor the existing system was takingmore time to evaluate the answers compared to the proposed system, We tried to evaluate the answersin very less time and more accurately.

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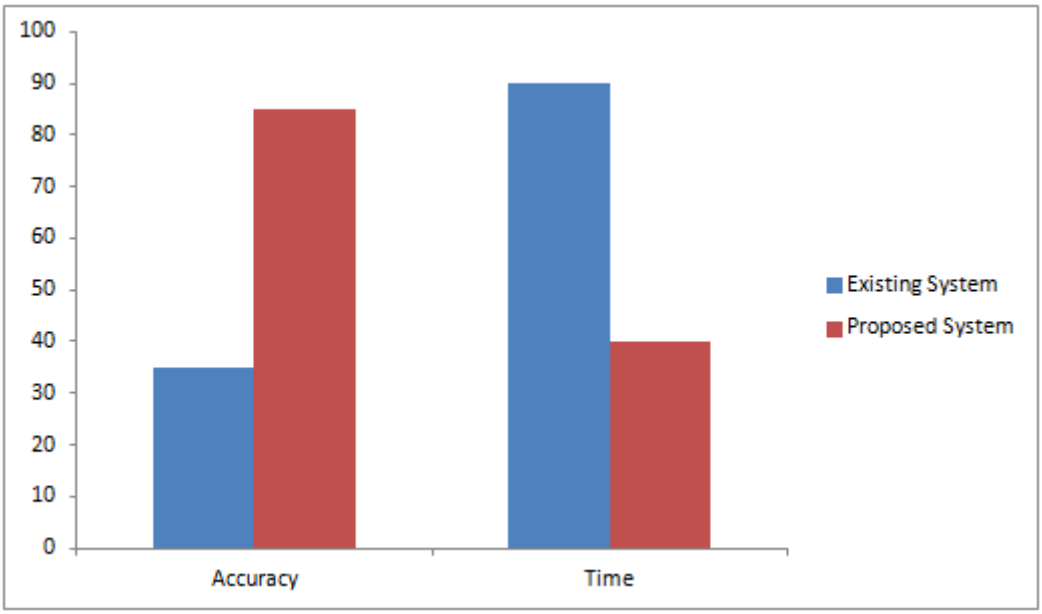

Figure 10: Bar Graph for Performance Evaluation

B. RESULTS

The proposed system results are achieved through Windows 10 Operating System with 64-bit configuration, intel CORE i3 processor and 4 GB RAM.

The below figure 11 represents the Home Page of AECF which includes student registration, login and Staff login.

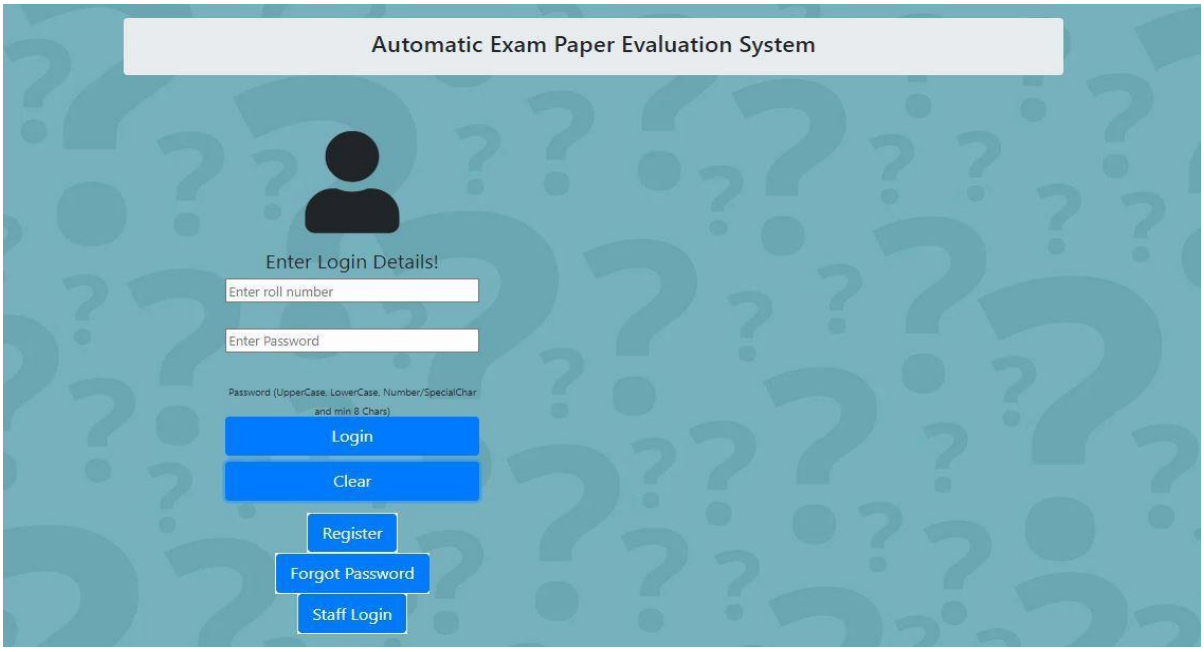

Figure 11: Home Page of AECF

The below figure 12 represents question and answer adding with maximum marks by evaluator.

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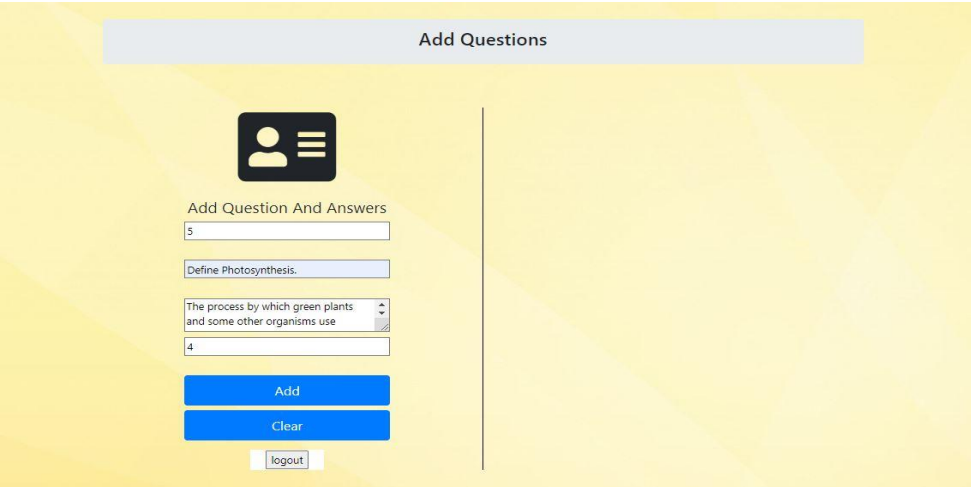

Figure 12: Adding Question and Key Answer by Evaluator

The below figure 13 represents the student written answer image uploading for the displayed question.

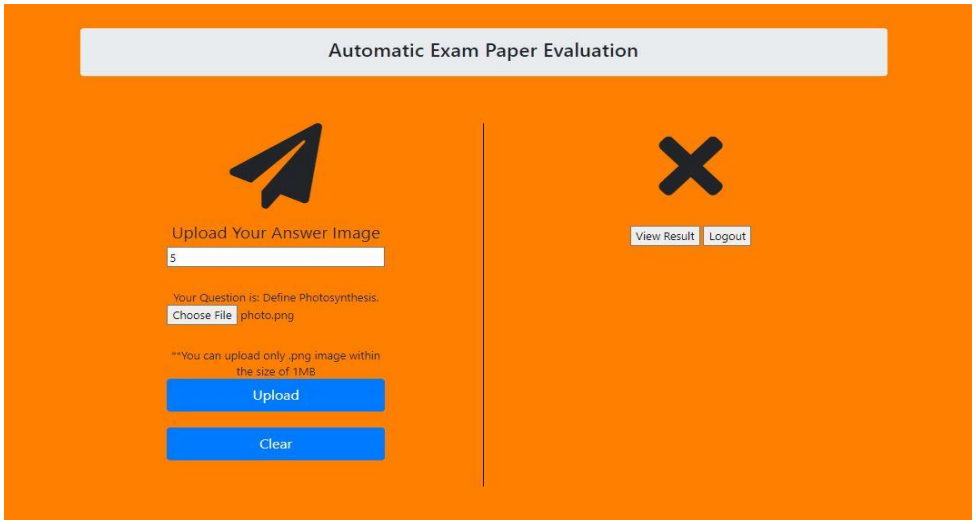

Figure 13: Selecting Question ID and Answer Uploading Page

The below figure 14 represents text extracted from answer image uploaded by student using OCR.

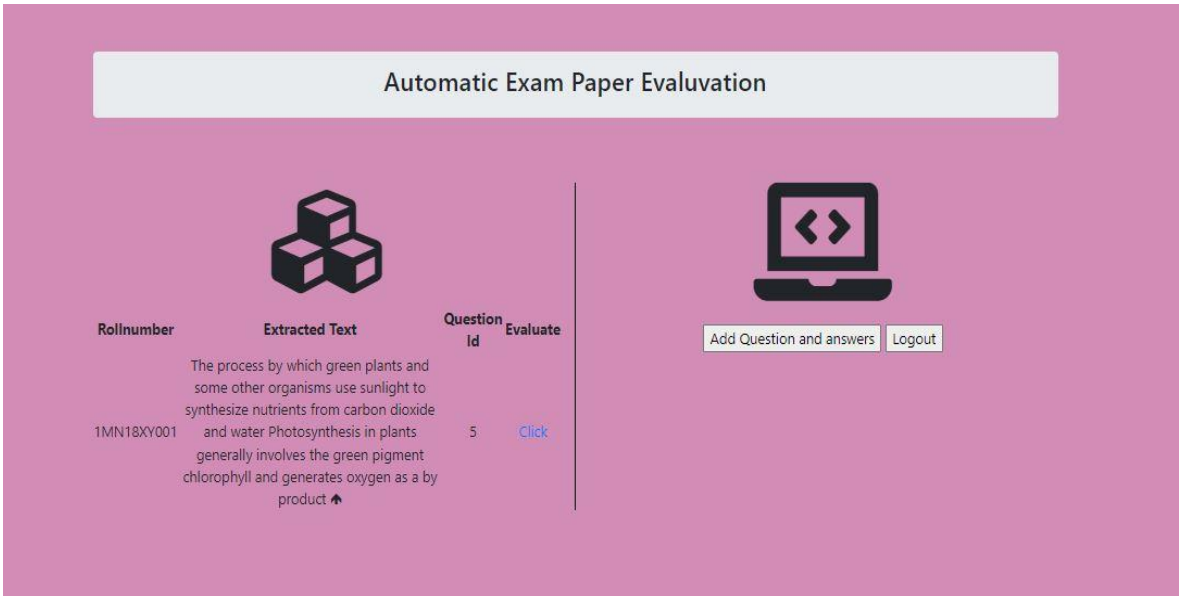

Figure 14: Displaying the OCR Extracted Text

The below figure 15 represents the marks displayed after evaluation.

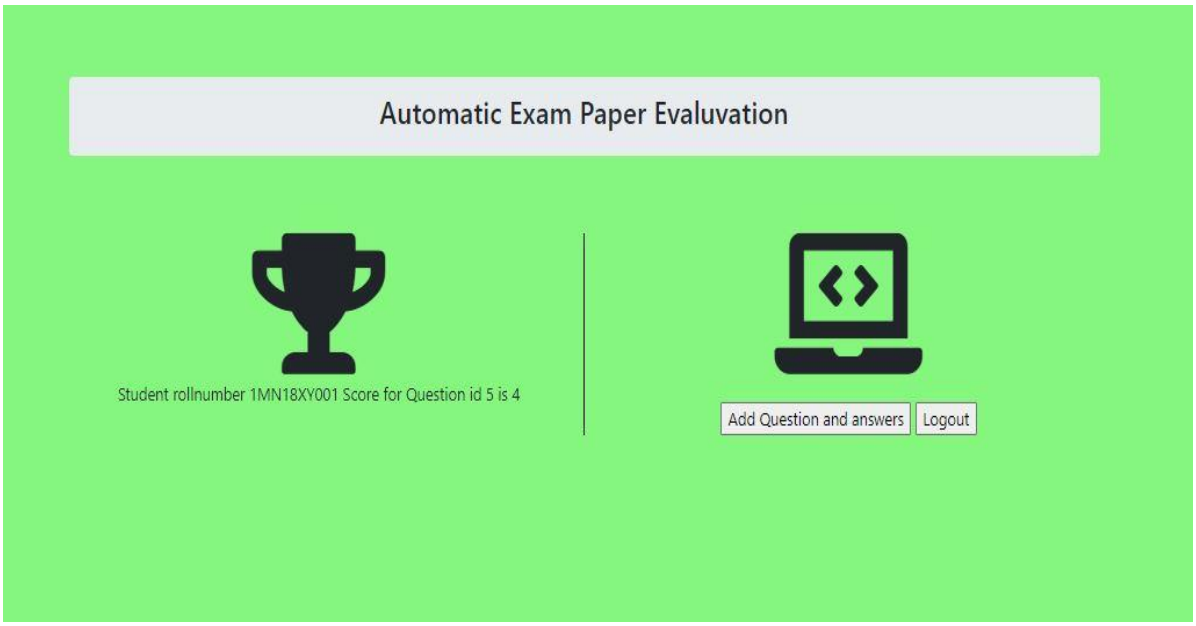

Figure 15: Marks Obtained

VII. CONCLUSION

All the studies which have been reviewed show that there are various different techniques for the evaluation of subjective answer sheets. The advantage of the system lies in the fact that it uses a weighted average of the closest to accurate techniques to provide the most optimized result. TESA is a systematic and reliable system which eases the role of evaluators and provides faster and more efficient outputs. This system offers a reliable, robust, and obvious short response time result. In the future, a system can be developed to evaluate diagrams as well as tables, an inbuilt system can also be made to type and make diagrams to shift examinations from handwritten paper based to completely online.

All the studies which have been reviewed show that there are various different techniques for the evaluation of subjective short answer sheets. As of now system uses weighted average of the closest to accurate technique to provide the most optimized result. It can be enhanced in future for correcting long answers. This system offers a reliable, robust, and obvious short response time result.

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